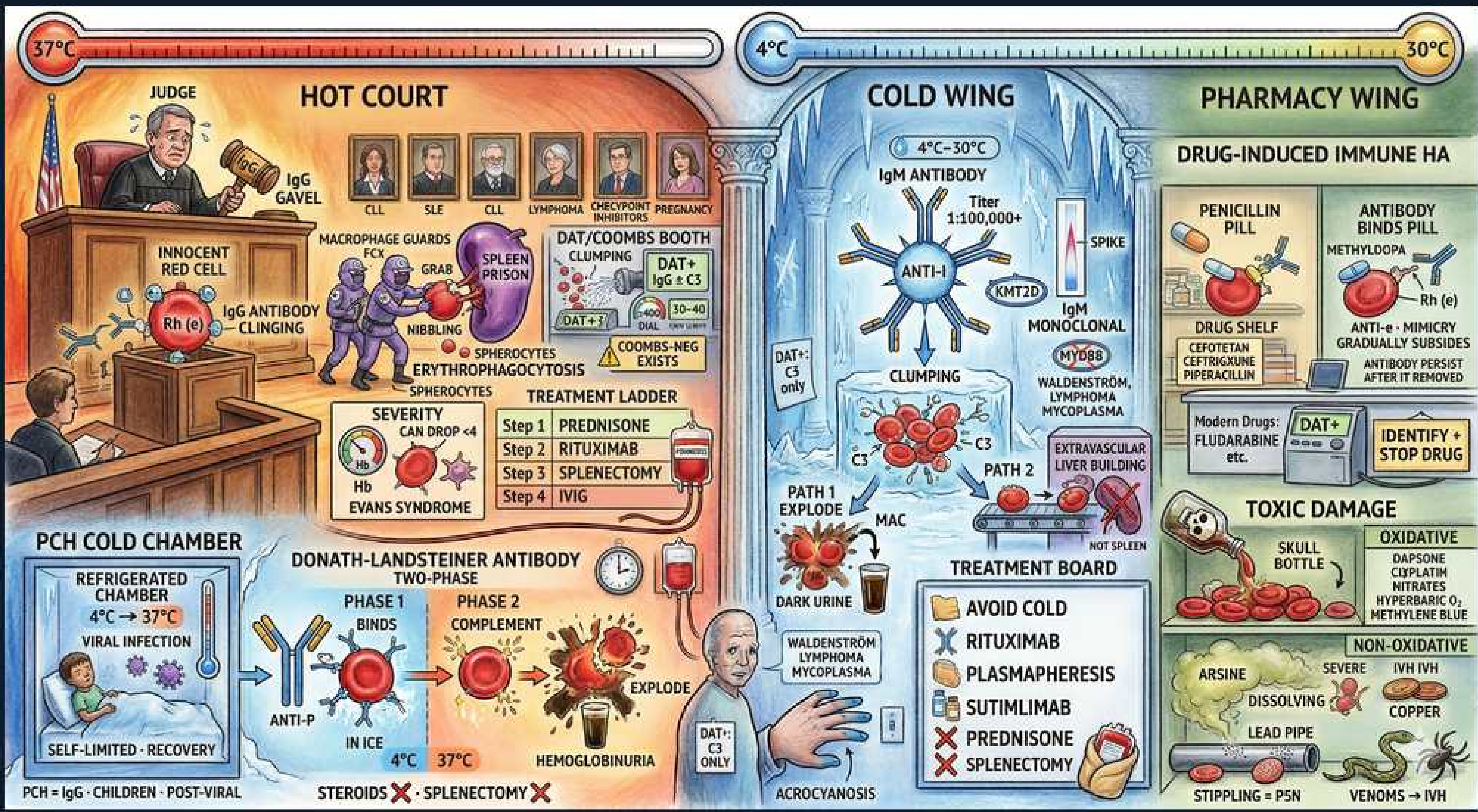
Autoimmune Hemolytic Anemia: Warm vs Cold



1. Warm AIHA = 37°C IgG

WHAT YOU SEE
Hot court, 37°C thermometer

WHAT IT ENCODES
Warm AIHA: IgG antibodies bind RBCs optimally at body temperature (37°C). Most common AIHA type. Extravascular hemolysis — splenic macrophages partially phagocytose IgG-coated cells producing spherocytes. DAT positive for IgG (+/- C3).

BOARD TRAP
Expecting intravascular hemoglobinuria in warm AIHA; it is mainly extravascular splenic clearance producing spherocytes.

2. IgG + spherocytes

WHAT YOU SEE
Macrophage nibbling RBC, spherocytes

WHAT IT ENCODES
Splenic macrophages bite off IgG/membrane, converting biconcave cells into spherocytes — the hallmark warm AIHA smear finding (also seen in HS, but DAT separates them). Reticulocytosis, high LDH, high indirect bilirubin accompany it.

BOARD TRAP
Calling spherocytes diagnostic of hereditary spherocytosis; in warm AIHA the DAT is positive — DAT distinguishes them.

3. Warm AIHA causes

WHAT YOU SEE
CLL, SLE, lymphoma, pregnancy guards

WHAT IT ENCODES
Warm AIHA is often secondary: CLL/lymphoma, SLE and other autoimmune disease, drugs, pregnancy; or primary/idiopathic. Evans syndrome = warm AIHA + immune thrombocytopenia. Always search for an underlying lymphoproliferative or autoimmune cause.

BOARD TRAP
Stopping at idiopathic; warm AIHA is frequently secondary to CLL, lymphoma, or SLE that must be excluded.

4. DAT / Coombs booth

WHAT YOU SEE
Coombs booth, clumping, DAT+

WHAT IT ENCODES
The direct antiglobulin test (DAT/Coombs) confirms immune hemolysis: warm AIHA is DAT+ for IgG (and/or C3); cold agglutinin disease is DAT+ for C3 only. A rare DAT-negative AIHA exists. DAT is the diagnostic centerpiece.

BOARD TRAP
Excluding AIHA on a negative DAT; rare DAT-negative AIHA exists when antibody load is below assay threshold.

5. Warm treatment ladder

WHAT YOU SEE
Steps — prednisone, rituximab, splenectomy, IVIG

WHAT IT ENCODES
Warm AIHA first line: CORTICOSTEROIDS (prednisone). Second line: rituximab or splenectomy. IVIG and immunosuppressants for refractory cases. Treat underlying cause. Steroids work because hemolysis is antibody-mediated and splenic.

BOARD TRAP
Using steroids/splenectomy for COLD agglutinin disease; those help warm AIHA but are ineffective for cold disease.

6. Cold agglutinin = 4°C IgM

WHAT YOU SEE
Cold wing, 4°C, IgM antibody

WHAT IT ENCODES
Cold agglutinin disease: IgM antibodies bind RBCs at cold temperatures (0-4°C, peripheral), fix complement, and cause complement-mediated (often intravascular) hemolysis plus agglutination. DAT positive for C3 only (IgM washes off). Acrocyanosis on cold exposure.

BOARD TRAP
Expecting IgG and a positive IgG DAT in cold disease; it is IgM-driven with a C3-only DAT.

7. Anti-I vs anti-i

WHAT YOU SEE
Anti-I, monoclonal vs anti-i

WHAT IT ENCODES
Anti-I specificity: chronic cold agglutinin disease (often monoclonal, MGUS/Waldenström) and MYCOPLASMA pneumoniae. Anti-i specificity: infectious mononucleosis (EBV). High thermal amplitude titers (1:100,000) drive hemolysis.

BOARD TRAP
Assigning mycoplasma to anti-i; mycoplasma and lymphoproliferative cold disease are anti-I, EBV mono is anti-i.

8. Cold associations

WHAT YOU SEE
Waldenström, lymphoma, mycoplasma

WHAT IT ENCODES
Cold agglutinin disease links to Mycoplasma pneumoniae, EBV, and lymphoproliferative disorders (Waldenström macroglobulinemia, lymphoma). Workup for an underlying B-cell clone is essential in chronic cases. Often monoclonal IgM.

BOARD TRAP
Missing the lymphoproliferative workup; chronic cold agglutinin disease is frequently due to an underlying B-cell clone.

9. Cold treatment board

WHAT YOU SEE

Avoid cold, rituximab, plasmapheresis; steroids/splenectomy X

WHAT IT ENCODES

Cold agglutinin disease: AVOID COLD (keep warm) is first; rituximab (+/- bendamustine), sutimlimab (complement C1s inhibitor), plasmapheresis for acute crises. Steroids and splenectomy DON'T work because hemolysis is complement/liver-mediated, not splenic.

BOARD TRAP

Giving prednisone/splenectomy for cold agglutinin disease; they fail — IgM-coated cells are cleared by complement in the liver.

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10. PCH cold chamber

WHAT YOU SEE

Refrigerated chamber, viral infection, kids

WHAT IT ENCODES

Paroxysmal cold hemoglobinuria (PCH): Donath-Landsteiner IgG biphasic antibody (anti-P) binds RBCs in cold, fixes complement, lyses on rewarming. Causes acute intravascular hemolysis with hemoglobinuria, classically in children post-viral. Usually self-limited.

BOARD TRAP

Confusing PCH with cold agglutinin disease; PCH is an IgG biphasic Donath-Landsteiner antibody (anti-P), not IgM cold agglutinin.

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11. Donath-Landsteiner test

WHAT YOU SEE

Two-phase test, binds in ice, explodes warm

WHAT IT ENCODES

The Donath-Landsteiner test confirms PCH: antibody binds RBCs at 4°C (phase 1) then complement lyses them on warming to 37°C (phase 2) — a biphasic hemolysin. This biphasic behavior is unique to PCH.

BOARD TRAP

Using a routine cold agglutinin titer for PCH; PCH needs the specific biphasic Donath-Landsteiner test.

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12. Drug-induced immune HA

WHAT YOU SEE

Pharmacy wing, penicillin, methyldopa

WHAT IT ENCODES

Drug-induced immune hemolysis: hapten (penicillin, high-dose, drug-on-cell), autoantibody (methyldopa — antibody persists after stopping), and immune-complex/innocent bystander (some cephalosporins). Identify and STOP the offending drug.

BOARD TRAP

Continuing methyldopa because antibody is present; methyldopa induces a true autoantibody — stop the drug, hemolysis resolves.

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13. Stop the drug

WHAT YOU SEE

Identify + stop drug

WHAT IT ENCODES

The key step in drug-induced immune hemolysis is identifying and discontinuing the culprit; most cases resolve after withdrawal. DAT may stay positive for weeks (especially methyldopa type). Supportive care; steroids occasionally.

BOARD TRAP

Reaching for immunosuppression first; drug withdrawal is the definitive treatment for drug-induced immune hemolysis.

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14. Oxidative drug damage

WHAT YOU SEE

Skull bottle, oxidative — dapsone

WHAT IT ENCODES

Distinguish immune drug hemolysis from oxidative (non-immune) drug hemolysis: dapsone, oxidizing agents cause Heinz-body/bite-cell hemolysis (DAT negative), prominent in G6PD deficiency. Mechanism (immune vs oxidative) changes the DAT and smear.

BOARD TRAP

Lumping all drug hemolysis as immune; oxidative drug hemolysis (dapsone) is DAT-negative with bite cells, not antibody-mediated.

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15. Severity / Hb drop

WHAT YOU SEE

Severity gauge, Evans syndrome

WHAT IT ENCODES

AIHA can cause rapid severe anemia; in Evans syndrome it coexists with immune thrombocytopenia (and sometimes neutropenia). Monitor Hb closely; transfusion may be needed despite crossmatch difficulty from autoantibody.

BOARD TRAP

Withholding transfusion in life-threatening AIHA because of incompatible crossmatch; give least-incompatible blood when Hb is critical.

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16. Acrocyanosis

WHAT YOU SEE

Blue fingers, cold-exposed extremities

WHAT IT ENCODES

Cold agglutinin disease causes acrocyanosis/Raynaud-like bluish discoloration of cold-exposed extremities (ears, nose, fingers) from agglutination in cool peripheral blood, reversible on warming. A bedside clue to cold-type AIHA.

BOARD TRAP

Mistaking acrocyanosis for true ischemia; cold agglutinin acrocyanosis reverses with rewarming, unlike arterial occlusion.
